Campus Shuttle Bus Simulation

How bus frequency changes queue length and waiting time

Monte Carlo simulation • 10,000 runs

Simulation focus

01 Reproduce the morning bus queue

02 Model uncertainty in arrivals and operations

03 Compare bus_interval = 4, 5, and 6 min

MATLAB implementation: main.m / run_single_simulation.m / plot_simulation_results.m

1. Simulation Overview

Morning shuttle queue from 7:00 to 10:00

INPUT

Train arrivals

Students arrive in groups according to the train timetable.

→

STATE

Bus queue

Non-walking students join a FIFO queue with their arrival times stored.

→

OUTPUT

Bus service

Buses arrive with random delays and capacities, then board students.

Key assumptions

- 1 minute per slot
- Train-based group arrivals
- 10% choose to walk
- Bus capacity: 70–80
- Bus delay: 0–5 min

The model tracks each student's arrival minute, making both FIFO boarding and individual waiting-time calculation possible.

2. Three-File Architecture

Each file has one clear responsibility

01

main.m

Experiment controller

- Sets schedules and parameters
- Runs 10,000 simulations
- Aggregates means and standard deviations

→

02

run_single_simulation.m

Single-run engine

- Generates bus capacity and delay
- Processes train and bus events
- Calculates waiting-time metrics

→

03

plot_simulation_results.m

Visualization

- Receives the mean queue history
- Plots average queue length over time
- Keeps analysis and plotting separate

3. How the Queue Is Reproduced

A student-level FIFO model

1

Train arrives

A group size is generated from a normal distribution.

2

Walking choice

Each student independently walks with probability 0.10.

3

Join queue

Non-walking students are stored by arrival minute.

4

Bus arrives

Capacity and delay vary in every simulation run.

5

FIFO boarding

Earliest arrivals board first; others remain.

student_arrivals = [arrival_time_1, arrival_time_2, arrival_time_3, ...]

```
Wait time = bus arrival slot − student arrival slot
```

Why this matters

The model measures average wait, maximum wait, queue peaks, transported students, and remaining students.

4. Monte Carlo Simulation

Why the same morning is simulated 10,000 times

Random elements in one run

Student arrivals	Normal distribution
Walking choice	10% probability
Bus capacity	70–80 students
Bus delay	0–5 minutes

rng(1) fixes the random-number sequence, so comparisons are reproducible.

Run 1 → Run 2 → Run 10,000

↓

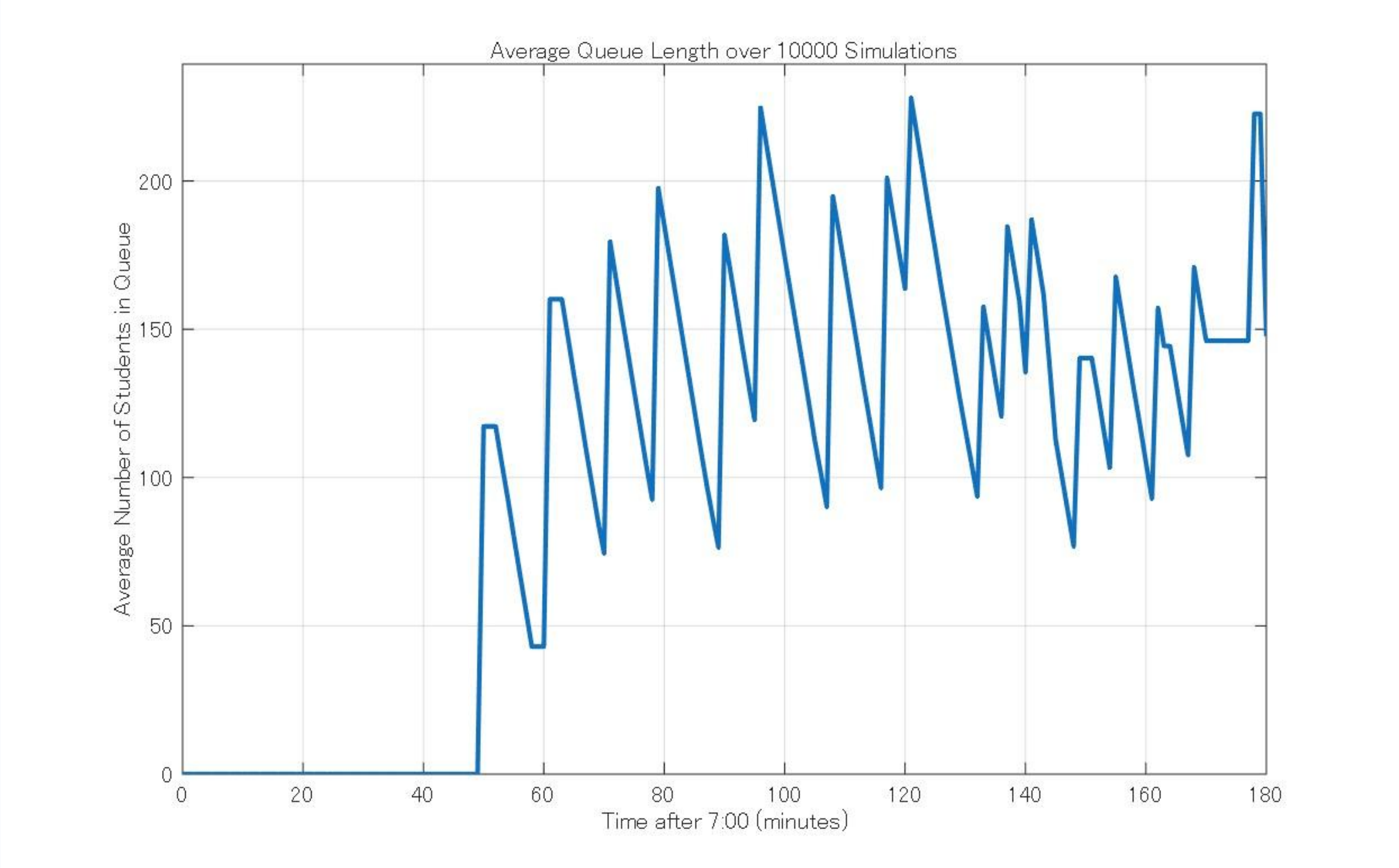
Average across all runs

Stable estimate of queue length and performance metrics

One simulation may be unusually good or bad. Repetition reveals the typical system behavior.

5. Baseline Result: bus_interval = 6 min

Average queue length over 10,000 simulations



Transported

1,478.11 students

Remaining

147.63 students

Average wait

11.42 min

Maximum wait

22.79 min

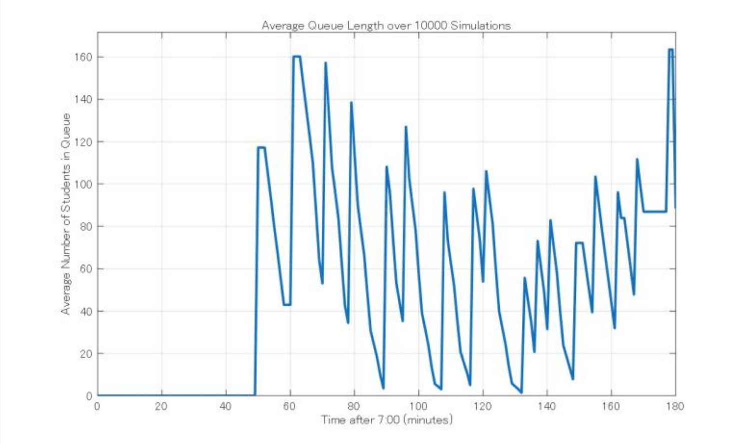
Maximum queue

270.52 students

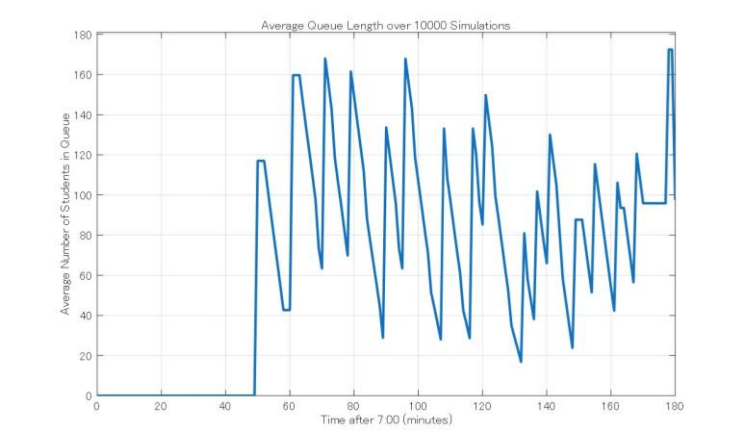
At 6-minute intervals, the system cannot fully clear the morning demand by 10:00.

6. Effect of Changing bus_interval

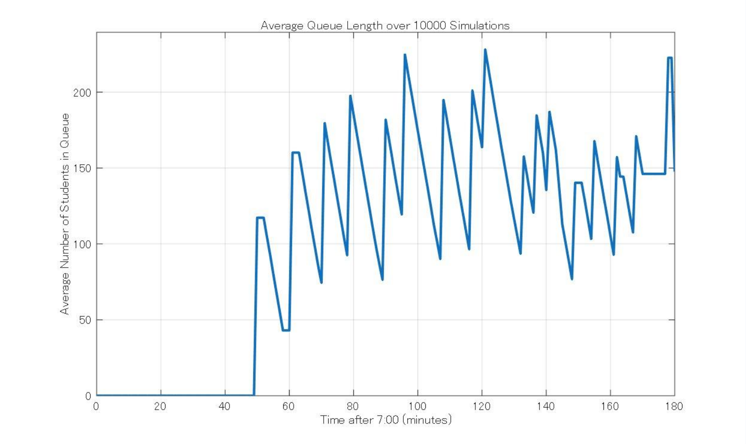
Shorter intervals reduce congestion and waiting time



4 min



5 min



6 min

Metric	4 min	5 min	6 min
Transported students	1,536.21	1,527.38	1,478.11
Remaining at 10:00	88.40	97.47	147.63
Average wait (min)	5.47	7.39	11.42
Maximum wait (min)	17.93	18.60	22.79
Maximum queue	200.67	219.72	270.52

Key finding

Changing 6 → 4 min cuts average waiting time by about 52% and reduces the maximum queue by about 26%.

7. Conclusion

Simulation supports data-driven bus scheduling

01 **The queue was reproduced realistically**

Train-based group arrivals, walking choice, FIFO boarding, random bus capacities, and random delays were included.

02 **Bus interval strongly affects performance**

As the interval increased from 4 to 6 minutes, queue length and waiting time increased clearly.

03 **Four-minute service performed best**

It transported more students, left fewer students waiting, and produced the shortest average wait among the tested settings.

Final assignment Group-B

- BRUNO ZOMIGNANI PERCIANI
- AOI TSUNEOKA
- RIN WAKAMATSU